Riya Prasad

Embedded Firmware Engineer

RESUME_1672

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Total Experience: 4 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 4 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Oscilloscope, C, SPI, UART, STM32, Embedded Linux, JTAG, C++, CAN, FreeRTOS, Firmware Debugging, ARM Cortex-M

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | PrismArc Solutions | Jan 2024 - Present

- Validated 676 hardware test vectors across protocol and timing corner cases
- Lowered field-reset incidents by 19% with watchdog recovery and diagnostics
- Improved battery runtime by 21% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | Ironleaf Engineering | Jul 2022 - Dec 2023

- Improved battery runtime by 21% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 16% using interrupt priority and queue tuning

PROJECTS

Motor Control Interface | C++, Firmware Debugging, JTAG, UART

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | SPI, ARM Cortex-M, C, CAN

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Shivanta College of Technology, 2022
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2023)